

2024 ANNUAL REPORT

# Cloudflare Radar

## 2024 Year in Review

Global Internet Traffic · Technology Adoption · Security Landscape

Data-driven analysis from Cloudflare's global network infrastructure

**+17%**

Traffic Growth YoY

**225**

Major Disruptions

**6.5%**

Traffic Mitigated

# 17% Global Traffic Growth and Starlink's 3.3× Surge

Global Internet traffic grew 17% YoY in 2024; Starlink's 3.3× surge signals satellite connectivity is entering the mainstream.



## +17%

Year-over-year global Internet traffic growth

Driven by commerce, comms & entertainment



## 3.3×

SpaceX Starlink traffic surge in 2024

Expanding to underserved regions globally



## 41%

Global traffic from mobile devices

Reflecting continued mobile-first trends

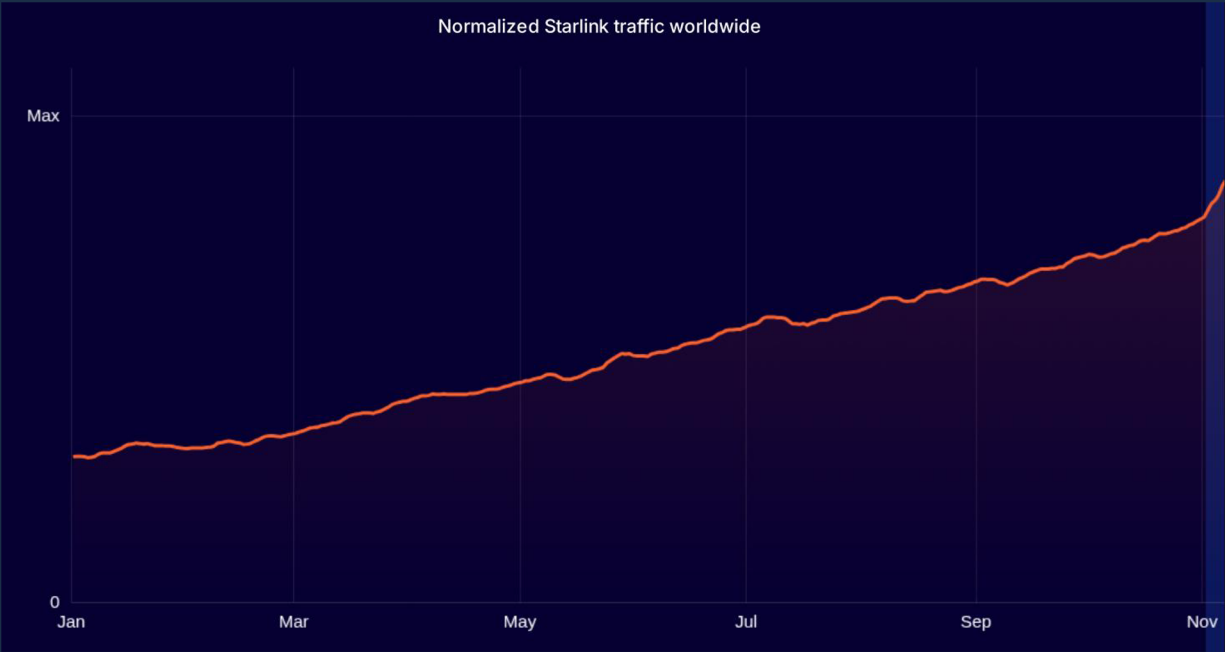
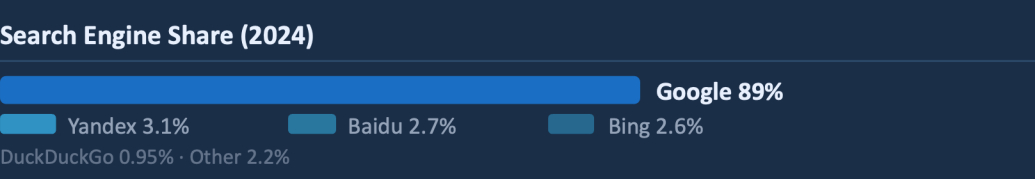


## 28%

IPv6 dual-stack adoption

Spain recorded highest download speed

Trend measured from mid-January to normalize post-holiday activity.  
Starlink scaled relative to peak observed request volume (not globally available).



### Starlink Traffic Growth — 2024

Normalized worldwide traffic showing 3.3× YoY increase  
Bringing satellite Internet to previously underserved locations

# Google Leads Top 10 Services as Generative AI Rises

Google topped rankings for the 2nd consecutive year; Generative AI was 2024's fastest-rising category while Metaverse faded.

## TOP 10 MOST POPULAR INTERNET SERVICES — 2024

| RANK | SERVICE   | CATEGORY             |
|------|-----------|----------------------|
| #1   | Google    | Search / Cloud       |
| #2   | Facebook  | Social Media         |
| #3   | Apple     | Hardware / Services  |
| #4   | TikTok    | Short-form Video     |
| #5   | AWS       | Cloud Infrastructure |
| #6   | Microsoft | Cloud / Productivity |
| #7   | Instagram | Social Media         |
| #8   | YouTube   | Video Streaming      |
| #9   | Amazon    | E-commerce / Cloud   |
| #10  | WhatsApp  | Messaging            |

Rankings based on Cloudflare 1.1.1.1 DNS resolver aggregate data



### Google #1 for the 2nd Consecutive Year

Maintained top position across all Cloudflare 1.1.1.1 DNS resolver aggregate traffic metrics in 2024



### Generative AI: Fastest-Rising Category

Continued rapid ascent throughout 2024 — ChatGPT, Claude, Gemini driving category growth



### Metaverse Receded into Background

VR/AR-focused services showed declining share of DNS traffic throughout the year

| Rank | Service   |
|------|-----------|
| #1   | Google    |
| #2   | Facebook  |
| #3   | Apple     |
| #4   | TikTok    |
| #5   | AWS       |
| #6   | Microsoft |
| #7   | Instagram |
| #8   | YouTube   |
| #9   | Amazon    |
| #10  | WhatsApp  |

Top 10 Internet Services Worldwide — Cloudflare Radar 2024

# Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

Browser defaults are accelerating enterprise encryption transitions — PQE adoption jumped 6× in months; HTTP/3 hits 21%; Go leads API traffic.

 **POST-QUANTUM ENCRYPTION · TLS 1.3**

2%

Early 2024

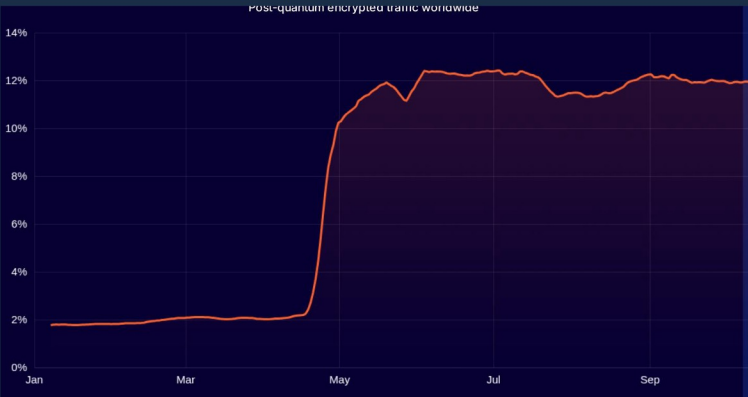
→

13%

End 2024

Chrome 124 default (Apr 17) triggered rapid adoption

Firefox rolling out; Safari testing in progress



Post-quantum encrypted traffic worldwide — surge following Chrome 124 release (April 17, 2024)

## RELATED TECHNOLOGY ADOPTION METRICS

 **HTTP/3 Adoption**

21%

of global traffic uses HTTP/3

QUIC-based: faster connections, packet-loss mitigation, default TLS

*Most recent HTTP protocol version*

 **Go Tops API Client Languages**

~12%

of automated API requests

Ranked: Go ~12% · Python 9.6%

Java 7.4% · .NET 3.6%

*Go surpassed Node.js for top spot*

 **Mobile OS & Browser Share**

33%

of traffic from iOS devices

Android peak >95% (some regions)

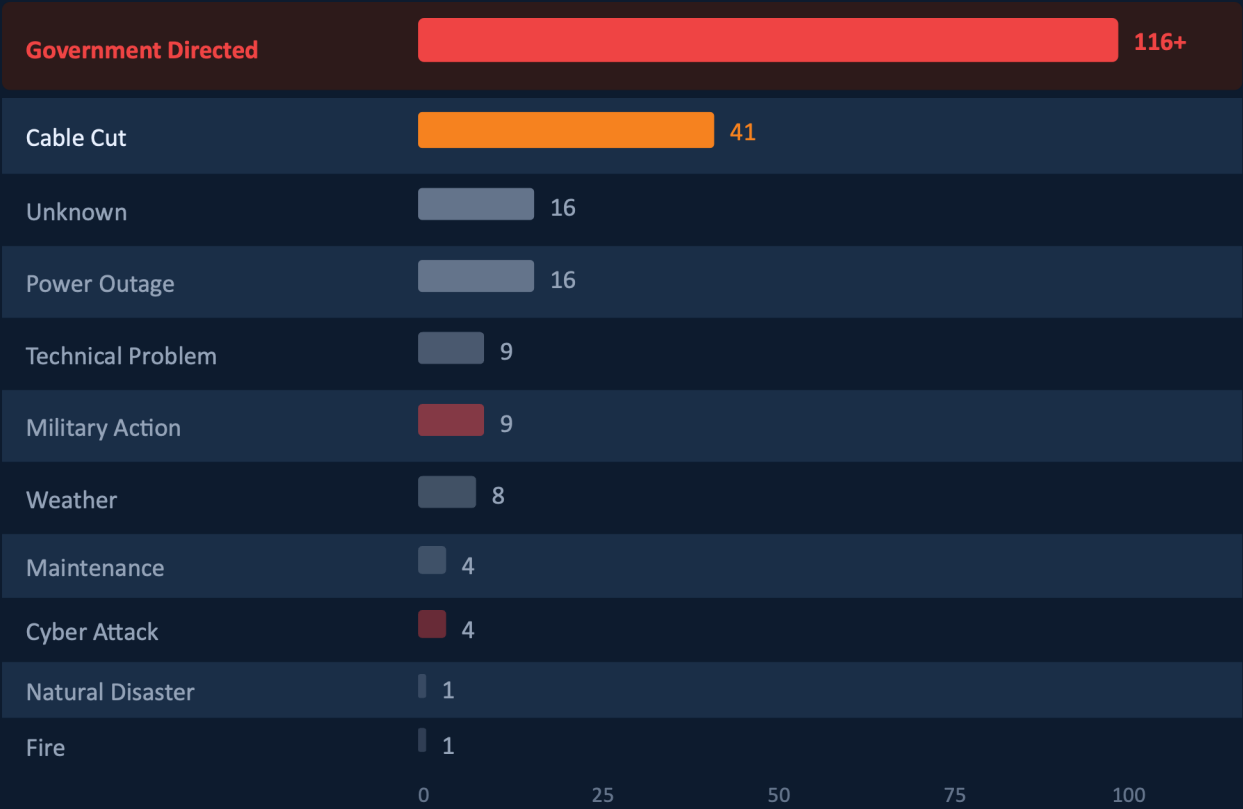
iOS peak ~66% · Chrome leads globally

*iOS + Android dominate mobile ecosystem*

# 225 Major Disruptions: Government Shutdowns Lead Outage Causes

Government-directed shutdowns (116+) account for over half of all 2024 Internet disruptions — dwarfing all technical causes combined.

## DISRUPTIONS BY CAUSE (2024 TOTAL: 225)

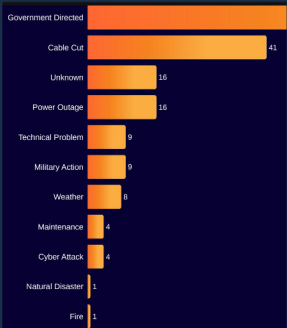


**225** Major Internet Disruptions globally observed in 2024  
Tracked by Cloudflare Radar Outage Center

**Government-Directed: 116+ incidents**  
Leading cause — over 51% of all disruptions  
Intentional state-directed Internet shutdowns

**Cable Cuts: 41 incidents**  
Second largest cause — physical infrastructure failures  
Undersea and terrestrial cable damage

**Power & Unknown: 16 each**  
Power outages and unattributed causes tied  
Military action: 9 · Weather: 8 · Cyber Attack: 4



Cloudflare Radar — Internet Disruption Causes 2024

# 6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5% traffic mitigation and Log4j's persistence 3+ years post-disclosure confirm that the threat surface is broad, constant, and slow to close.



### TRAFFIC MITIGATED

6.5%

of all Cloudflare-processed traffic



### BOT TRAFFIC CONCENTRATION

68%

from just top 10 countries



### ROUTING SECURITY (RPKI)

+6.4%

RPKI valid routes increase in 2024

## Threat Landscape



### Most Attacked Sector

Gambling & Games

Highest attack volume industry category observed



### Log4j Still Actively Exploited

CVE-2021-44228 — disclosed December 2021

Continues to be targeted 3+ years after initial disclosure

*Underscores persistent nature of widely known vulnerabilities*



### Bot Traffic Geographic Concentration

68% of global bot traffic from top 10 countries

Highly concentrated origin geography for automated attacks

### Strategic Implication:

Continuous patching, zero-trust architecture, and behavioral threat intelligence are non-negotiable in 2025

## Email Security Threats



4.3%  
of emails processed  
were malicious

Identified via Cloudflare email security analysis

43% contained a deceptive link

Designed to trick recipients — most common malicious payload  
Of all malicious emails processed

Top Malicious TLD: .bar

Largest share of malicious and spam email origin domain  
Monitor unconventional TLDs in email filtering policies

### Strategic Implication:

Advanced email security with link analysis and TLD-based reputation scoring is critical for enterprise threat reduction

# Five Takeaways Shaping the 2025 Internet Outlook

2024's data reveals an Internet growing faster, encrypting quicker, fracturing politically, and under relentless threat — demand action in 2025.

01



**Traffic Growth Trajectory**

**+17% YoY**

Starlink 3.3× surge

Satellite connectivity enters the mainstream for underserved regions

**2025 Action**  
Plan infrastructure capacity for accelerating demand

02



**Post-Quantum Encryption**

**2% → 13%**

of TLS 1.3 traffic

Chrome 124 default triggered rapid jump  
Firefox + Safari following

**2025 Action**  
Accelerate PQC migration planning — window narrowing

03



**Connectivity Fragility**

**116+**

gov't shutdowns

225 total disruptions  
Shutdowns now #1  
outage cause globally

**2025 Action**  
Build multi-path connectivity for at-risk regions

04



**Persistent Security Threats**

**6.5%**

traffic mitigated

Log4j exploited 3+ yrs  
4.3% malicious email  
Gambling #1 targeted

**2025 Action**  
Defense-in-depth: patch, email sec, DDoS mitigation

05



**Developer Ecosystem Shift**

**Go #1**

API client language

HTTP/3 at 21%  
Protocol stack evolving rapidly in 2024

**2025 Action**  
Align API gateway and CDN to modern protocol stacks